

Kelly Xu

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Software Engineer with industry internships delivering backend, full-stack, and cloud solutions.
Striving to make tech more reliable, empathetic, and accessible.

EDUCATION

Duke University, Durham, NC – B.S. in Computer Science, Concentration in Software Systems

GPA: 3.82, Expected May 2026

Involvement: DTech Scholar (2023-present) · Software Developer, Duke CFCI (2024-2025) · Researcher, Duke Database Research Group (2025-present) · Director, Duke Applied Machine Learning (2024–2025)

Coursework: Operating Systems, Database Systems, Computer Architecture, Computer Network Architecture, Software Engineering, Web Development, Design and Analysis of Algorithms, Machine Learning.

Teaching Assistant | Duke, Durham NC

- Data Structures and Algorithms (CS201) Aug 2024 - present
- Introduction to Database Systems (CS316) Aug 2025 - present

SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, HTML/CSS

Frameworks & Tools: React, Node.js, Express.js, Flask, PostgreSQL, AWS, Docker, CircleCI

INTERNSHIPS

Full Stack Software Engineer Intern | Carter's Inc, Atlanta GA

May 2025 - Aug 2025

- Refactored and modernized AWS Infrastructure-as-Code and CI/CD pipelines for Carter's loyalty platform using AWS CDK, TypeScript, and CircleCI, **reducing code duplication by ~35%, improving system maintainability by ~25%, and decreasing deployment time by ~15%** across Development, UAT, and Production environments.
- Developed scalable TypeScript APIs using AWS API Gateway and Lambda in an Agile team, improving product recommendations and sales across **1000+ Carter's stores and multi-million-user platform**.

UI/UX Design Intern | Merck, Rahway NJ

Jun 2024 - Aug 2024

- Developed an automation tool that reduced website content migration time by **90%**.
- Redesigned and rebuilt a critical internal website using Figma, HTML, and CSS, improving usability and efficiency **for 1,000+ users**.

Software Engineer Intern | Duke + CISCO, Durham NC

May 2023 - Aug 2023

- Developed a server activity logging application for phishing attack prevention using Python and socket programming; deployed on Duke's network to **mitigate 50 million daily phishing attacks**.
- Delivered client-configurable features and Dockerization for the STINGAR cybersecurity platform, collaborating with CISCO engineers to improve scalability and integration for enterprise use.

PROJECTS

SwimConnect Admin Dashboard | Python, Flask, SQLAlchemy, PostgreSQL, HTML/CSS

Jan 2025 - May 2025

- Developed a production-ready administrative dashboard for an athlete mentorship startup, delivering a client-ready MVP in 8 weeks to streamline operations for 50+ users.
- Led backend and database design with Flask and SQLAlchemy for a 5-person team, cutting manual tracking time by 90% with role-based admin features.

[Amazon Plant Marketplace](#) | React, JavaScript, HTML/CSS, Python/Flask, PostgreSQL

Sept 2024 - Dec 2024

- Designed and implemented a scalable PostgreSQL schema and RESTful API with Flask to support browsing, buying, selling, and reviews, collaborating in a team of 5 students.
- Optimized performance with indexing and pagination to handle 9,000+ product listings efficiently.